Year 9, Chapter 4.4: Lenses and the Eye

What is a lens?

_____

_____

A ***convex*** lens _____ outwards. These lenses cause the light rays to _____ together, or converge. If a convex lens is held close to an object, it can be used as a _____ glass. It produces an upright and _____ virtual image.

Comparison of convex lenses

Weak flattened lens	Strong curved lens

Describe the following terms:

Focal length: (label on above diagrams)

_____

_____

Real image:

_____

_____

Draw a ray diagram (see instructions on page 135, fig 4.4.5), for when a **real inverted image** is produced.

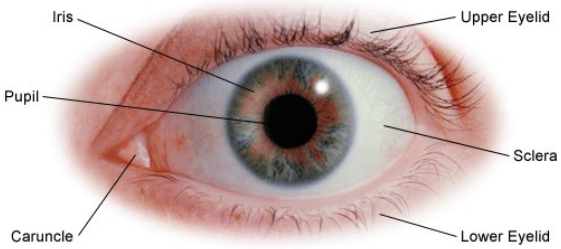
A ***concave*** lens curves _____. These lenses cause light rays to _____ or spread out. They spread lights rays as though they have come from a point _____ the lens. A concave lens produces images that are _____, _____ and virtual.

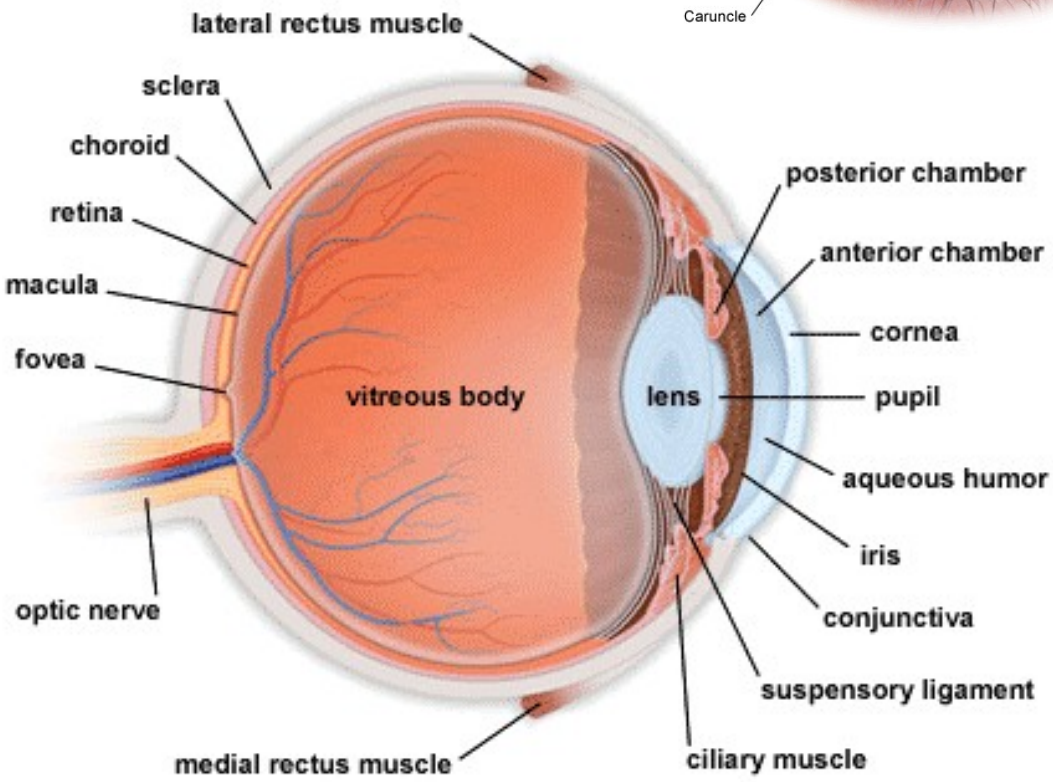
Draw a labelled diagram of a ***concave*** lens

The Eye

Light enters the eye and is refracted by the _____, and focused by the lens, which is _____. Incoming light is actually focused to form a clear, upside-down image on the _____, at the back of the eye. This image is converted into a series of electrical signals, which then travel along the _____ _____ to the brain. The brain then interprets this information as an image.

Diagram of the Eye





Task: Describe the role of each of the structures of the eye

(Use you textbooks, pg 137, to add any unlabelled structures to the diagrams above)

Aqueous humour

_____

_____

Cornea

_____

_____

Lens

_____

_____

Ciliary muscle/body

_____

Conjunctiva

Choroid

Sclera

Fovea (centralis)

Optic nerve

Eye muscles

Vitreous humour

Iris

Pupil

Sphincter muscle

_____

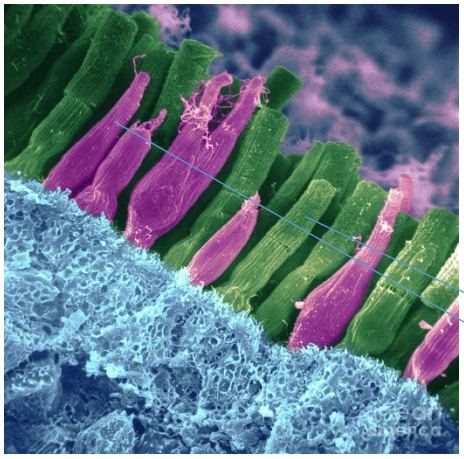
_____

Dilator muscle

_____

_____

Rods and Cones



Cone cells are more concentrated (packed) in the _____ and are receptive to _____.

Rod cells are located around the _____ and are nerve cells sensitive to _____ light.

_____

_____

Vision problems (*pg 138*)

The lenses in your eyes focus on objects at different distances by changing focal length (remember focal length = distance from lens to focus).

Describe the action of the muscles attached to lens when a person is looking at an object:

that is far/distant

_____

_____

that is close by

_____

_____

What is 'accommodation'?

_____

Why does accommodation get harder as we age?

_____

What is myopia?

With reference to lens type, how can myopia be corrected?

What is hyperopia?

With reference to lens type, how can hyperopia be corrected?

Application!!!! The Bionic Eye!

Using page 139, briefly bullet point key ideas as to how the bionic eye aids with impaired vision.